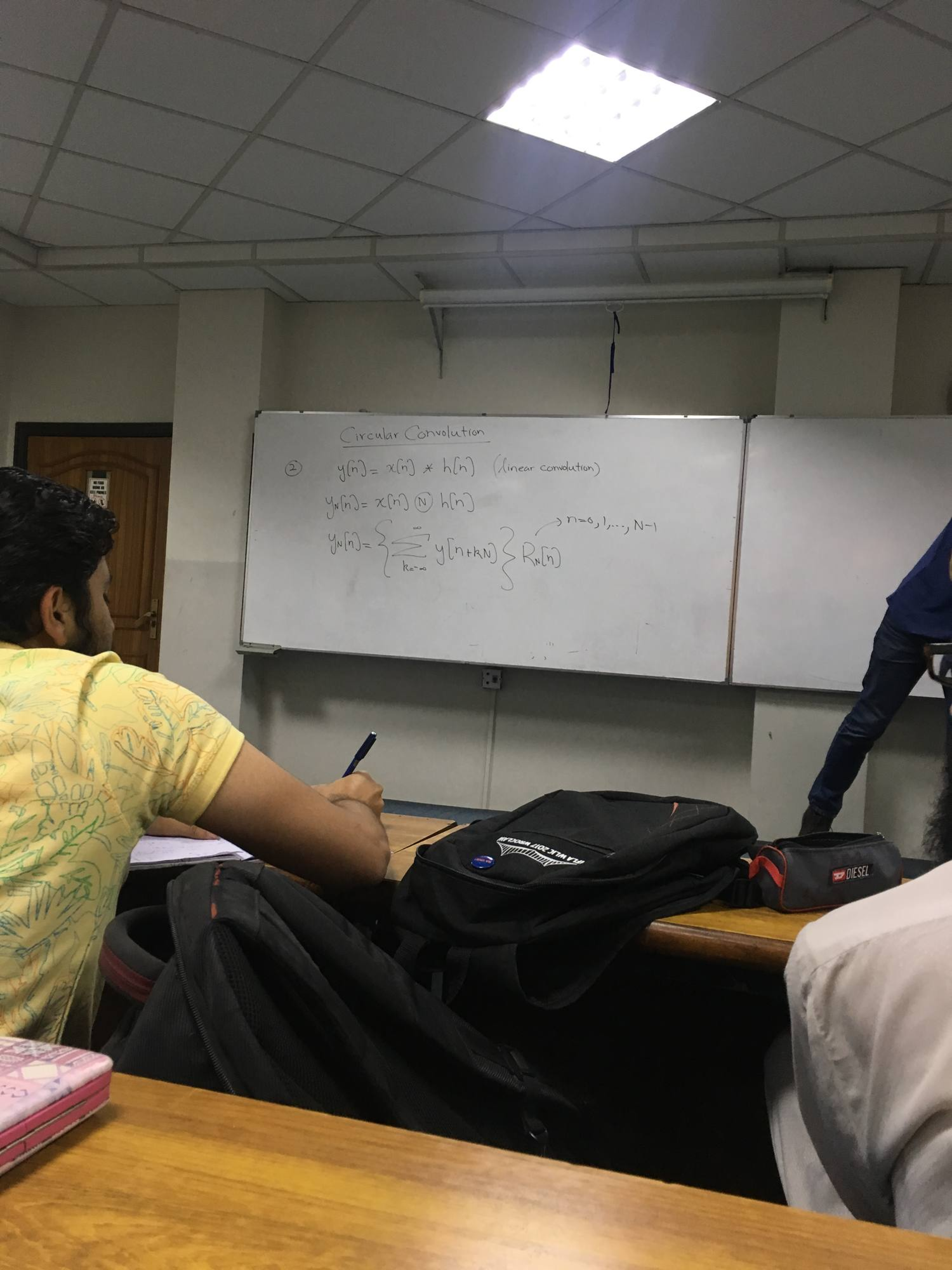
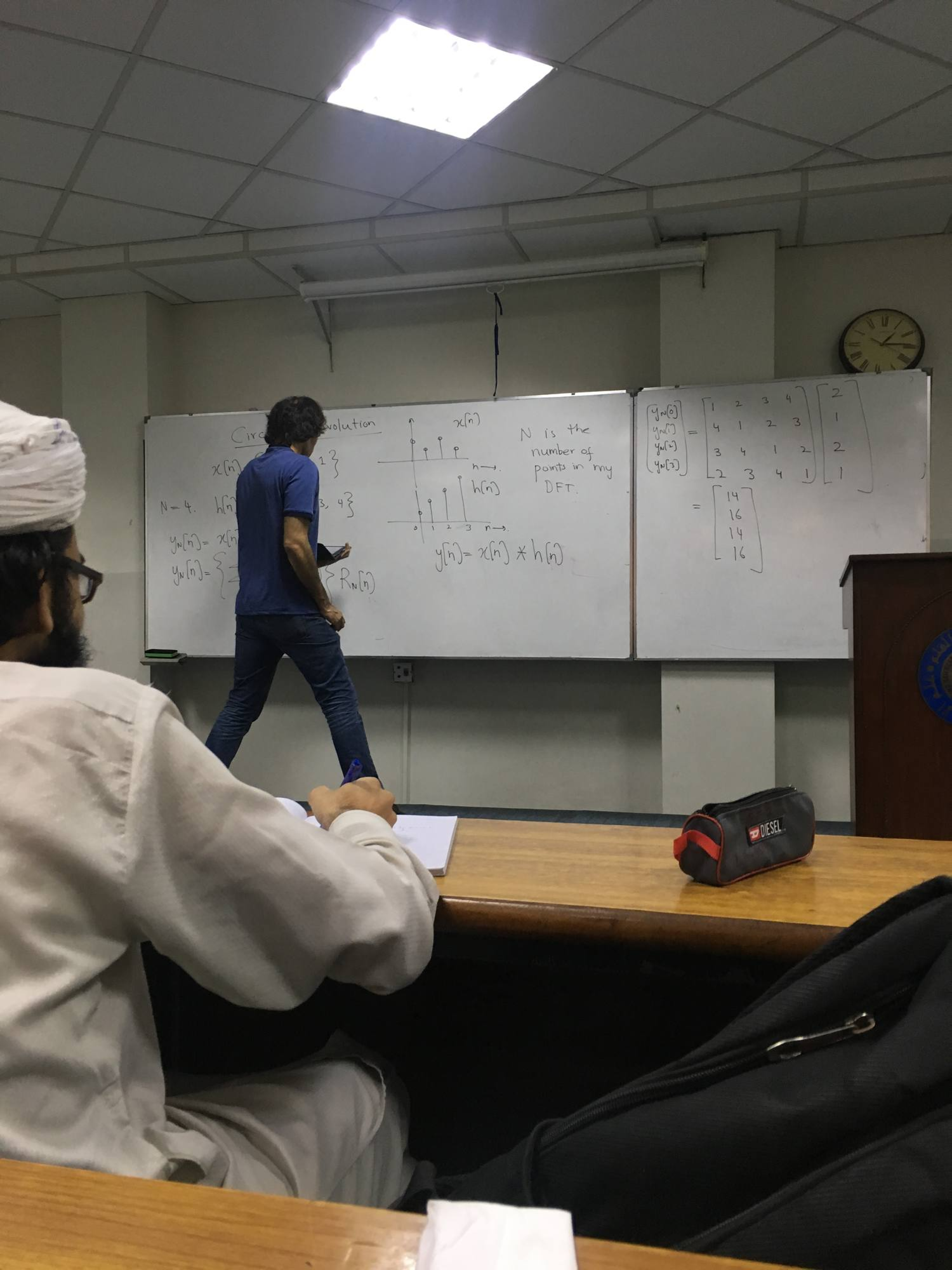
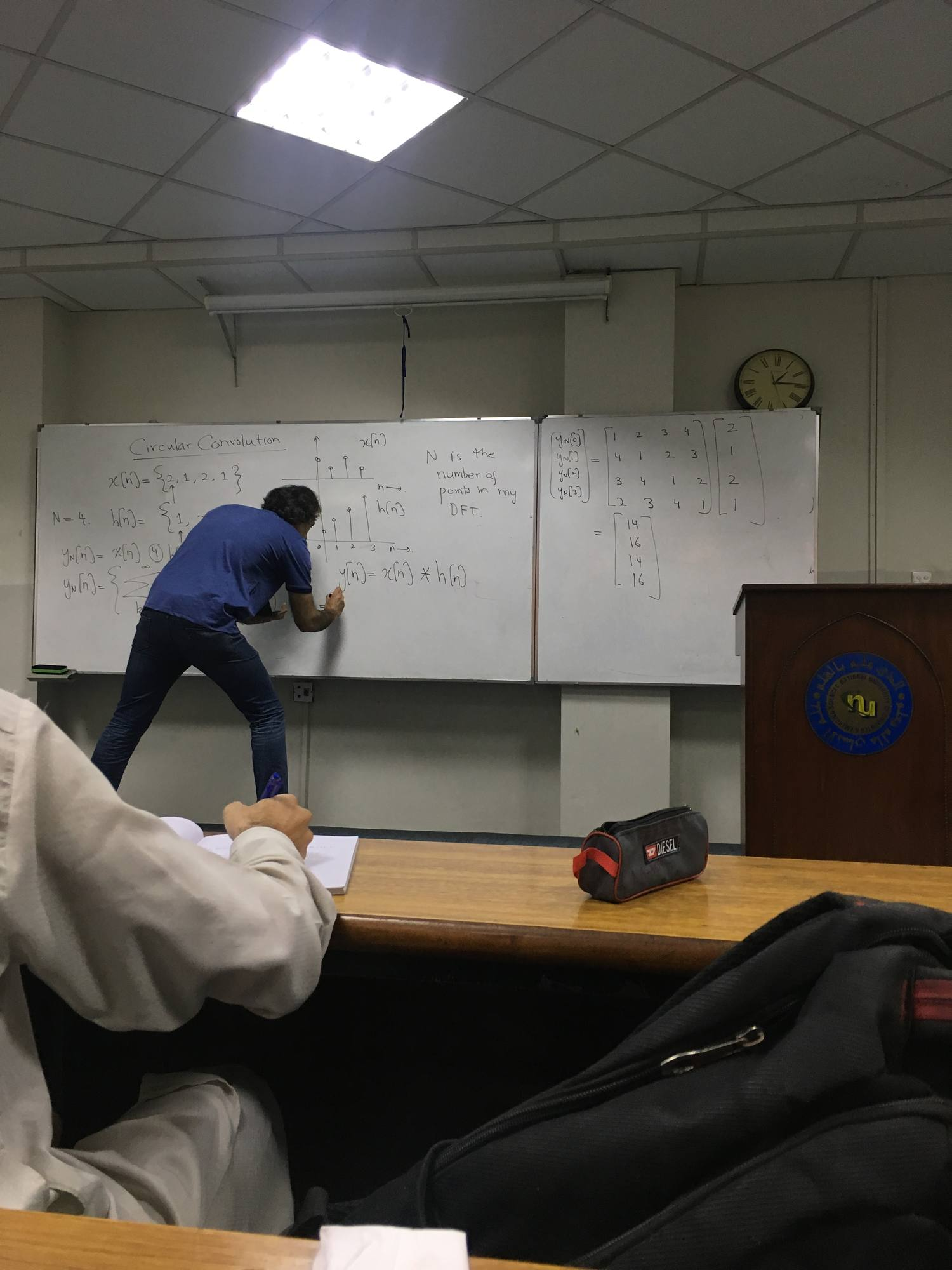


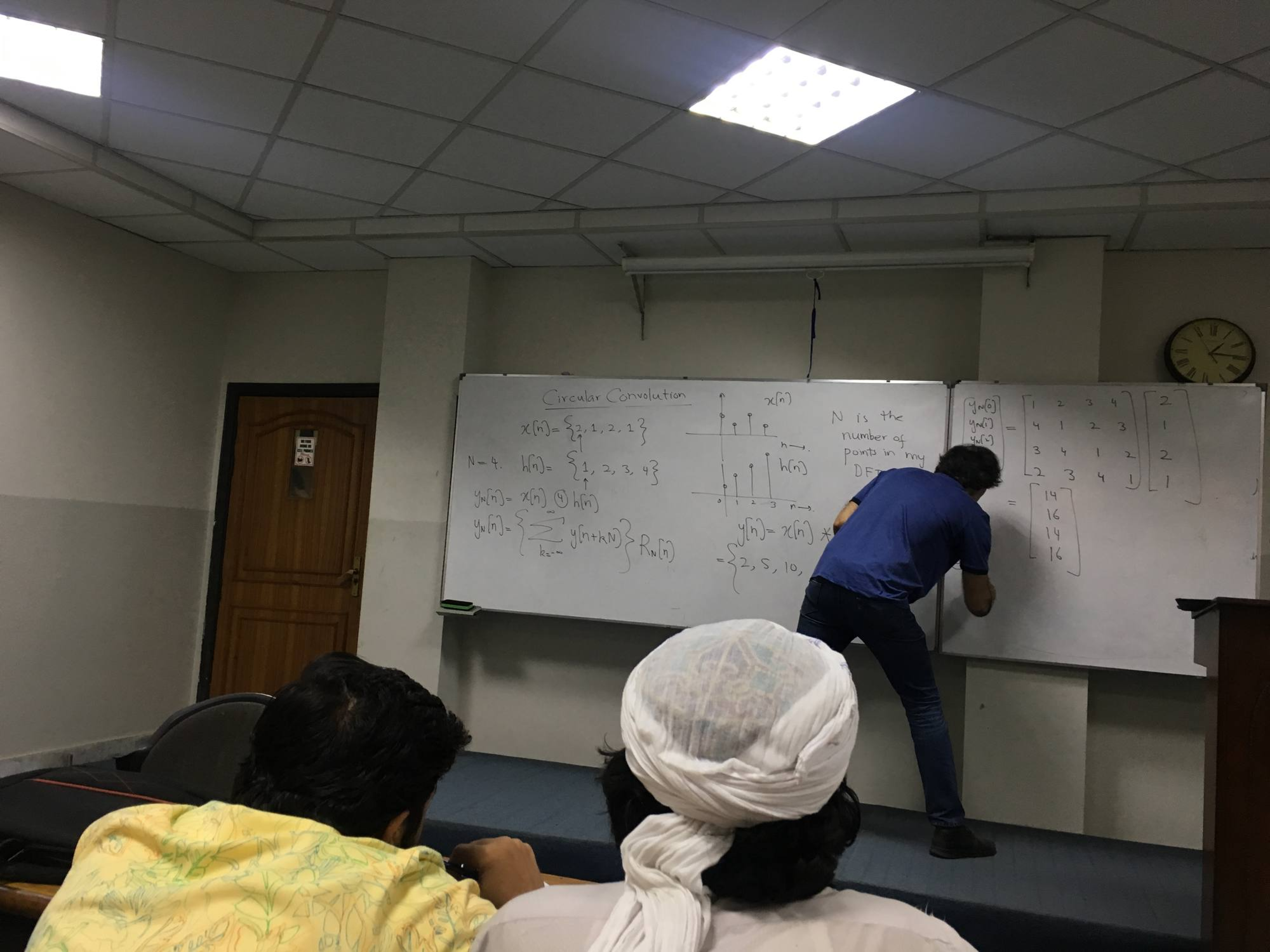
Circular Convolution
(2) $y[n] = x[n] * h[n]$ (linear convolution)
$y_N[n] = x[n] \, (N) \, h[n]$
$y_N[n] = \left\{ \sum_{k=-\infty}^{\infty} y[n+kN] \right\} R_N[n]$
$n = 0, 1, \ldots, N-1$
DIESEL

nvolution
x(n)
N is the number of points in my DFT.
$R_N[n]$
$\begin{bmatrix} y_N[0] \\ y_N[1] \\ y_N[2] \\ y_N[3] \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \end{bmatrix}$
$= \begin{bmatrix} 14 \\ 16 \\ 14 \\ 16 \end{bmatrix}$
DIESEL

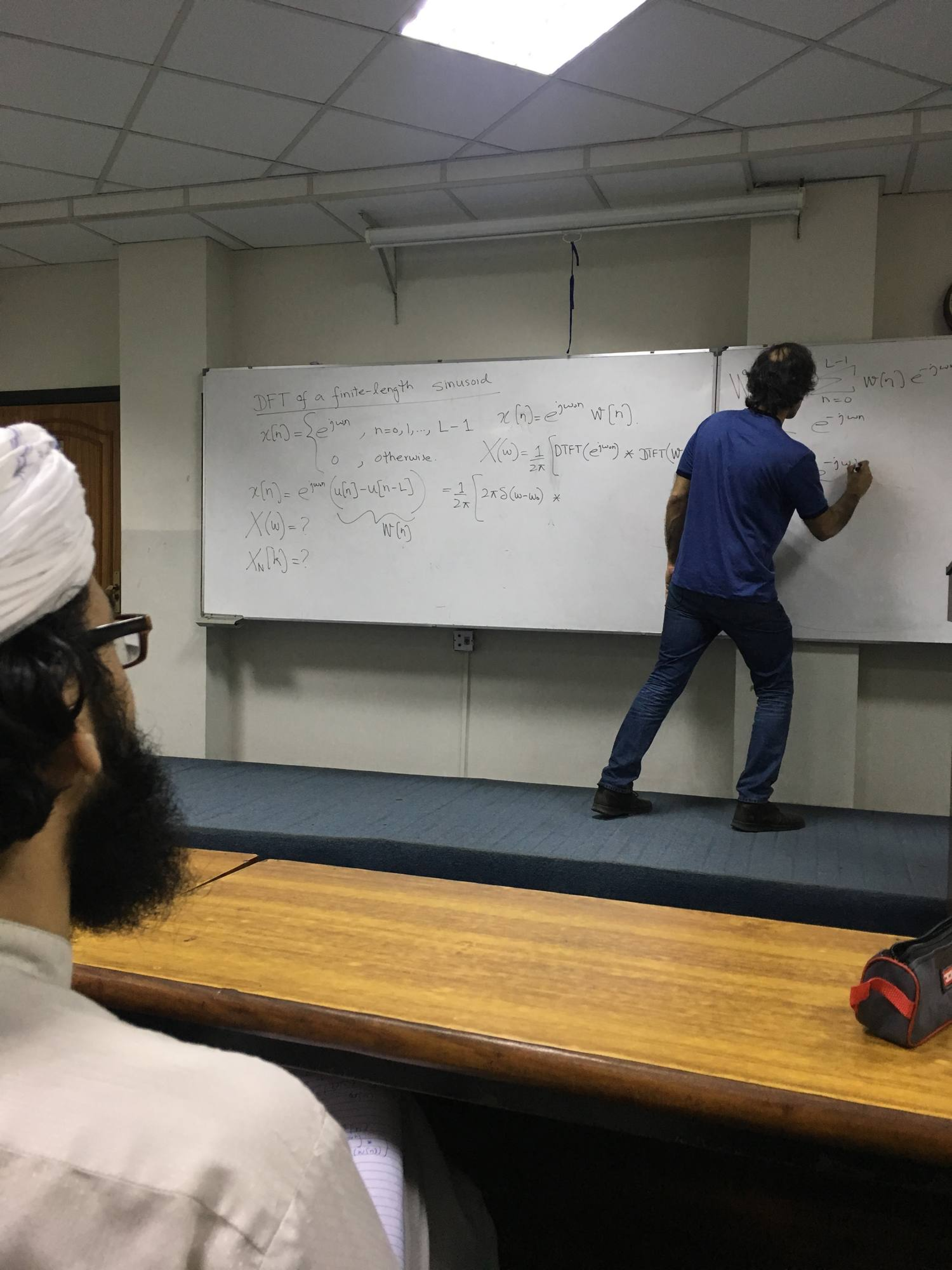

x(n)
h(n)
n⟶
N is the number of points in my DFT.
N = 4.
y(n) = x(n) * h(n)
R_N(n)
y_N(0)
y_N(1)
y_N(2)
y_N(3)
1 2 3 4
4 1 2 3
3 4 1 2
2 3 4 1
2
1
2
1
14
16
14
16
DIESEL



Circular Convolution
x(n) = {2, 1, 2, 1}
N = 4.
h(n) =
x(n)
h(n)
N is the number of points in my DFT.
y(n) = x(n) * h(n)
yN(0)
yN(1)
yN(2)
yN(3)
1 2 3 4
4 1 2 3
3 4 1 2
2 3 4 1
2
1
2
1
14
16
14
16
DIESEL



Circular Convolution
x[n] = {2, 1, 2, 1}
N = 4. h[n] = {1, 2, 3, 4}
y_N[n] = x[n] 4 h[n]
y_N[n] = { Σ y[n+kN] } R_N[n]
k=-∞
x(n)
h(n)
n→
y[n] = x[n] *
= {2, 5, 10,
N is the number of points in my
y_N[0]
y_N[1]
y_N[2]
1 2 3 4
4 1 2 3
3 4 1 2
2 3 4 1
2
1
2
1
14
16
14
16



DFT of a finite-length sinusoid
x[n] = { e^jω₀n , n=0,1,..., L-1
0 , otherwise.
x[n] = e^jω₀n w[n].
X(ω) = 1/2π [DTFT(e^jω₀n) * DTFT(W
x[n] = e^jω₀n (u[n] - u[n-L])
W[n]
= 1/2π [2πδ(ω-ω₀) *
X(ω) = ?
X_N[k] = ?
L-1
n=0
W(n) e^-jω
e^-jωn

of a finite-length sinusoid

$$= \begin{cases} e^{j\omega n}, & n = 0, 1, \ldots, L-1 \\ 0, & \text{otherwise.} \end{cases}$$

$$x[n] = e^{j\omega_0 n}\, w[n]$$

$$X(\omega) = \frac{1}{2\pi}\left[\text{DTFT}(e^{j\omega_0 n}) * \text{DTFT}(w[n])\right]$$

$$= \frac{1}{2\pi}\left[2\pi\,\delta(\omega - \omega_0) * W(\omega)\right]$$

$= ?$

$= ?$

$$\underbrace{\qquad}_{w[n]}$$

$$W(\omega) = e^{-j\omega\left(\frac{L-1}{2}\right)}\left[\frac{\sin\left(\omega L/2\right)}{\sin\left(\omega/2\right)}\right]$$

$$X(\omega) = W(\omega - \omega_0)$$

$$= e^{-j(\omega-\omega_0)\left(\frac{L-1}{2}\right)}\,\frac{\sin\left((\omega-\omega_0)L/2\right)}{\sin\left(\frac{\omega-\omega_0}{2}\right)}$$

$$x[n] * \sum_{k=-\infty}^{\infty} \delta[n-kN] \xleftrightarrow{\text{DTFT}} X(\omega) \sum_{k=-\infty}^{\infty} \frac{2\pi}{N}\delta\left(\omega-\frac{2\pi k}{N}\right)$$

$$\sum_{l=-\infty}^{\infty} x[n-lN] = \boxed{\frac{1}{N}\sum_{k=0}^{N-1} X_N[k]\, e^{j\frac{2\pi k}{N}n}}$$

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X_N[k]\, e^{j\frac{2\pi k}{N}n}$$

$N \geq L$

$n=0,1,\ldots,N-1$

$x(t) \xleftrightarrow{\text{CTFT}} X(F)$

$x[n] \xleftrightarrow{\text{DTFT}} X(\omega)$

If $N \geq L$ and $n=0,1,\ldots,N-1$, then

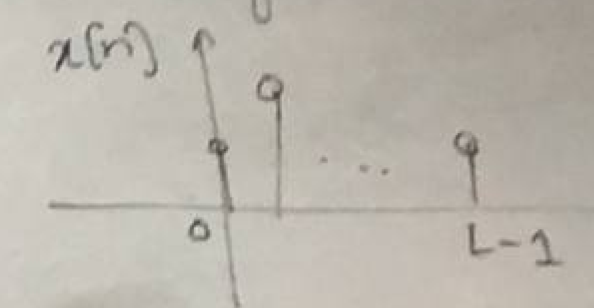
$$\sum_{l=-\infty}^{\infty} x[n-lN] = x[n]$$

for $n=0,1,\ldots,N-1$

$x[n]$ starts from t

and is of some f

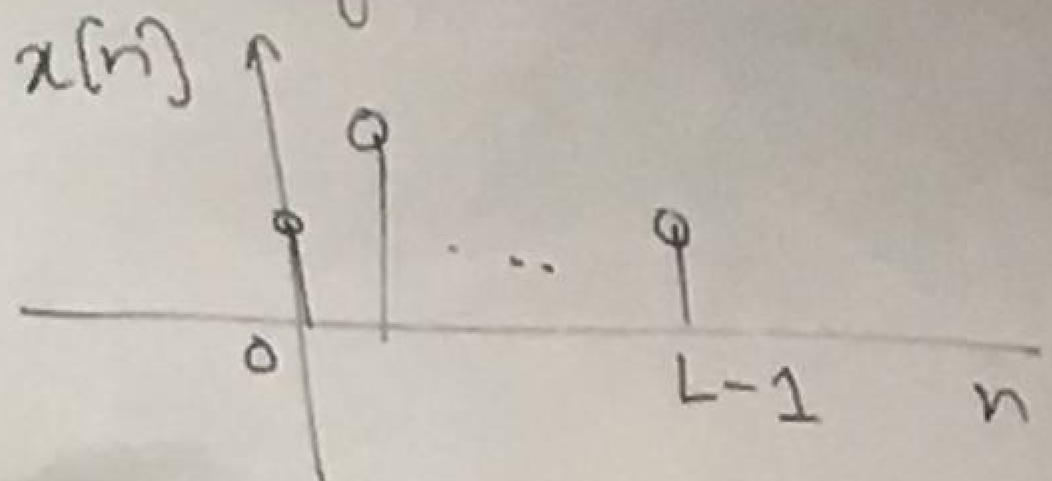
length L.



$x[n] \xrightarrow[N]{\text{DFT}} X$

$X_N[k] \xrightarrow[N]{\text{IDFT}} \sum$

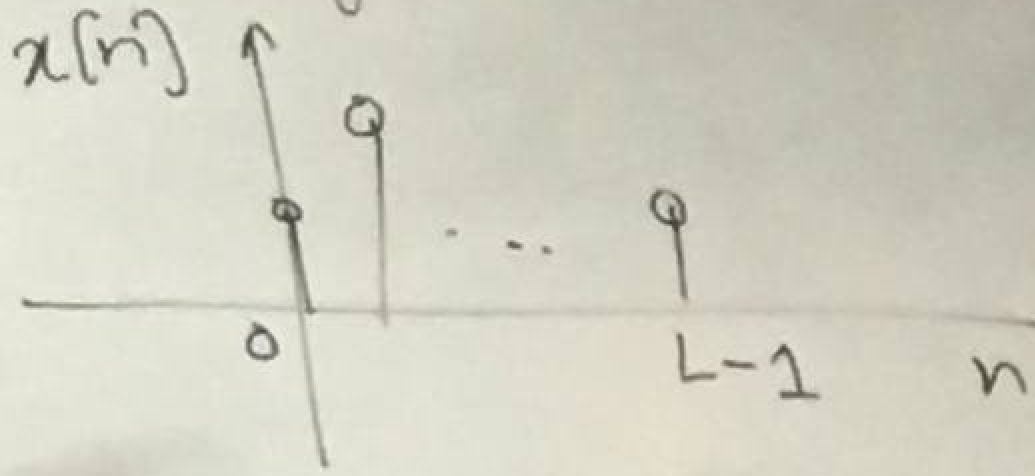
$x[n]$ starts from time zero and is of some finite length L.



$$x[n] \xrightarrow[N]{DFT} X_N[k]$$

$$X_N[k] \xrightarrow[N]{IDFT} \sum_{l=}$$

$x[n]$ starts from time zero and is of some finite length L.



$$x[n] \xrightarrow[N]{\text{DFT}} X_N[k]$$

$$X_N[k] \xrightarrow[N]{\text{IDFT}} \sum_{l=}$$